

Sajjad Hossain Talukder

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EDUCATION

Jan 2019 – Jan 2026 **Bachelor of Science in Computer Science & Engineering**
Premier University, Chattogram, Bangladesh
CGPA: 3.98 / 4.00 (1st rank in batch)

RESEARCH EXPERIENCE

2026 **PDAF — SDN-Assisted Adaptive Forwarding in NDN-MANETs**
Undergraduate thesis · Premier University, Dept. of CSE

- First-authored an SDN-assisted forwarding framework for NDN over MANETs — BFS-based node-disjoint route discovery, multi-metric route ranking, and network-wide packet suppression via an extended Dead Nonce List. Evaluated in ndnSIM/ns-3 against CSF, CBILEM and SDN-SRP — packet delivery ratio +25-86%, end-to-end latency -63-96%.

2026 – Present **FORCE — Federated ORchestration for Cache Eviction in Mobile NDN**
Joint research with Dr. S. M. A. Iqbal · Premier University, Dept. of CSE

- Building a state-aware cache-eviction meta-policy — a controller-hosted D3QN (double dueling DQN) with Prioritized Experience Replay selects each node's eviction policy from ~21 continuous state features, federated via a global Q-network, driven by a Python/ZeroMQ control plane bridged to ndnSIM.

PUBLICATIONS

PDAF: Performance-Driven Adaptive Forwarding in SDN-Assisted NDN-MANETs
S. H. Talukder, S. M. A. Iqbal, E. Barua
IEEE Transactions on Mobile Computing, 2026, in press, doi: 10.1109/TMC.2026.3722116

Empowering Bengali Language in Drone Control with Artificial Neural Networks
S. H. Talukder, N. Rezoana, T. Mahmud, N. Basnin, S. Chowdhury, M. S. Hossain, K. Andersson
Book chapter in Applied Machine Learning on Sensing Technologies, CRC Press / Taylor & Francis, pp. 81-93, 2026, doi: 10.1201/9781003506218-6

WORK EXPERIENCE

Nov 2025 – Present **Software Engineer**
Pivotly, Minnesota, USA (Remote)

- Build AI-powered backend services and agentic LLM workflows that automate high-volume business processes, develop the accompanying web and mobile frontends over REST APIs, and deploy them to production on cloud infrastructure.

Sep 2024 – Oct 2025 **Software Engineer**
Silicon Orchard Ltd., Dhaka, Bangladesh

- Built and maintained cross-platform (React Native) and native iOS (Swift) applications for global clients, integrating backend APIs and optimizing performance for reliability on unstable networks.

Mar 2024 – Sep 2024 **Software Engineer (iOS)**
Xotech, Chattogram, Bangladesh

- Developed AI-based iOS applications (Swift, UIKit) backed by PHP/Laravel REST APIs, managing the full lifecycle through to App Store release.

PROJECTS

Grade Now — AI Grading Platform for Students (Agentic AI, LLM, Node.js, Next.js, PostgreSQL, AWS)

- Solely built and deployed an AI platform that grades student answer scripts against AI-generated rubrics through a parallel, multi-stage agentic pipeline, cutting evaluation time and improving consistency.

- Connect My Advocate — Legal Aid Platform (Next.js, Node.js, PostgreSQL, Agora SDK, AWS)
 - Sole engineer behind the complete system — client web, mobile, and admin apps, plus the AWS infrastructure and PostgreSQL backend — for a legal-aid platform with real-time video consultation.
- IELTS Pro BD — Mock IELTS Test Platform (Next.js, Node.js, PostgreSQL, SSLCommerz)
 - Architected, shipped, and maintain a live computer-delivered mock-IELTS platform with student management, a flexible test-engine editor, and subscription billing.
- SmartWeight — Edge Vision Weighing System (Raspberry Pi, Python, OpenCV)
 - Integrated real-time OpenCV object detection with weight sensors on Raspberry Pi to automate billing and product identification, fully on-device.

AWARDS & ACHIEVEMENTS

Competitive Programming — ICPC	ICPC Asia West Championship Finalist, 2022 (Team PUC Eternals) ICPC Dhaka Regional — Main Round 52nd (2022), 78th (2023), 93rd (2021); Preliminary 76th of 2481 (2023)
Competitive Programming — Ratings	Codeforces: Specialist (max rating 1417), 1000+ problems solved CodeChef: 3-star (max rating 1721); also active on LeetCode, LightOJ and TOPH
Scholarships	Merit-Based Scholarship, Premier University

CERTIFICATIONS & TEST SCORES

IELTS Academic — Overall Band 7.0 (CEFR C1)
British Council / IDP · Listening 7.5, Reading 7.5, Writing 6.5, Speaking 6.0, 2026

SKILLS

Research & Simulation	ndnSIM, ns-3, Python/ZeroMQ control plane, D3QN + Prioritized Experience Replay, Federated Learning, Named Data Networking, SDN
Languages	JavaScript/TypeScript, Python, Swift, Go, C++, PHP, SQL
Cloud & DevOps	AWS, Azure, Docker, CI/CD (GitHub Actions), PostgreSQL, MySQL, Firebase, Linux
Frameworks & Backend	Next.js, React, Node.js, React Native, SwiftUI/UIKit, Laravel, RESTful APIs
AI & Systems	Agentic AI & LLM Integration, AI/ML, OpenCV, Deep Reinforcement Learning (D3QN, PER)

ACTIVITIES & LEADERSHIP

Jan 2021 – Dec 2023	Manager and Trainer (Competitive Programming) <i>Premier University Computer Club</i> Ran data-structures and algorithms bootcamps for competitive-programming teams.
Jan 2022 – Dec 2022	Program Secretary <i>Premier University Robotics Club</i> Ran programming classes for club members and arranged and managed the club’s events.

RESEARCH INTERESTS

Named Data Networking (NDN) & SDN, Deep & Federated Reinforcement Learning, In-Network Caching and Forwarding, Agentic AI, Quantum Computing

REFERENCES

- Dr. Shahid Md. Asif Iqbal (Undergraduate thesis supervisor)
Professor, Dept. of CSE; Associate Dean, Faculty of Engineering | Premier University, Chattogram | asif.iqbal@puc.ac.bd
- Amit Sarker (Hiring manager and team lead, Silicon Orchard 2024-25)
Software Engineer; formerly Lead Software Engineer, Silicon Orchard | Pivotly | amit@pivotly.com
- Mohammad Hasan (Moderator, Programming Wing, PU Computer Club)
Assistant Professor, Dept. of CSE | Premier University, Chattogram | mehedi.cse@puc.ac.bd